Wildlife Ecology

Week 4 Assignment Solutions (2026)

1 Consider the food chain: Grass → Grasshopper → Frog → Snake → Hawk. In this food chain,

a. hawk is producer

b. **hawk is consumer and carnivore**

c. hawk is consumer and herbivore

d. hawk is decomposer

2 Trees → Birds → Parasites → Hyperparasites represents

a. upright pyramid of numbers

b. **inverted pyramid of numbers**

c. spindle pyramid of numbers

d. dumb-bell pyramid of numbers

3 Glacial lakes are typical examples of

a. eutrophic lakes

b. hypereutrophic lakes

c. **oligotrophic lakes**

d. mesotrophic lakes

4 In a grazing food chain, the first trophic level after producers is generally occupied by:

(a) Carnivores

(b) Decomposers

(c) Detritivores

(d) **Herbivores**

5 Which statement correctly defines net primary production?

(a) Energy intercepted by producers from the Sun

(b) Total energy lost during transfer between trophic levels

(c) **Gross primary production minus energy lost through respiration per unit time**

(d) Energy stored only in herbivore biomass

6 Which statement correctly defines gross primary production?

(a) **Energy fixed through photosynthesis per unit time**

(b) Energy lost by organisms through respiration per unit time

(c) Total number of producers in an ecosystem

(d) Organic matter consumed by herbivores per unit time

7 Which chain best represents a detritus food chain?

(a) **Fallen leaves → detritivores → small fish → large fish**

(b) Grass → chital → tiger

(c) Rat → flea → parasitic protozoa

(d) Phytoplankton → zooplankton → fish → sea lion

8 Consider the food chain: Grass → Grasshopper → Frog → Snake → Hawk. As we move up the food chain,

a. **available energy decreases**

b. available energy increases

c. available energy remains same

d. available energy is zero everywhere

9 Which statement correctly describes a decomposer?

(a) It only feeds on living plants

(b) It consumes only primary consumers

(c) **It converts dead material into soil and recycles nutrients**

(d) It always occupies the quaternary trophic level

10 Which description best defines a heterotroph?

(a) An organism that converts sunlight directly into inorganic nutrients

(b) **An organism that cannot produce its own food and depends on organic carbon from other sources**

(c) An organism that uses chemical reactions to manufacture minerals

(d) An organism that occupies only the first trophic level